

NEPHROLOGY TEACHING SERIES

Dialysis & KRT

Indications, modalities, and access

Dialysis is started because of a problem the kidneys cannot solve — not because of a number.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why we get consulted

The consult is about a decision, not a diagnosis.

REASONS FOR KRT CONSULT

Approaching AEIOU (acidosis, electrolytes, ingestion, overload, uremia)

ESRD admitted, off-schedule, or missed

Refractory hyperK or pulmonary edema

Dialyzable toxin

CKD-5 — access + modality planning

PD peritonitis or catheter issue

CRRT vs IHD in the ICU

TEACHING PEARL

*State the problem, not the modality.
'Refractory hyperK despite therapy' is a reason. 'BUN 100' is not. Name the question, what failed, and the ask.*

AEIOU: the urgent indications

Any one of these → dialysis is not elective.



Acidosis

pH < 7.1–7.15 refractory to bicarbonate and ventilation.



Electrolytes

Life-threatening or refractory hyperkalemia: often $K^+ \geq 6.5$, or any dangerous ECG changes despite temporizing therapy.



Ingestion

Dialyzable toxin: methanol, ethylene glycol, salicylates, lithium, metformin (metformin-associated lactic acidosis (MALA)), valproate.



Overload

Refractory pulmonary edema; anuric CHF; need for space for blood products or TPN.



Uremia

Pericarditis, encephalopathy, bleeding diathesis, intractable nausea or pruritus.

STARRT-AKI / AKIKI: without AEIOU, watchful waiting is safe — ~40% avoid KRT entirely. Don't start for BUN/Cr alone.

Modalities at a glance

Hemodynamics and clinical context drive the choice.

	IHD	continuous renal replacement therapy (CRRT)	SLED	PD
Duration	3–5 h, 3×/wk	24 h continuous	8–12 h nightly	CAPD 4 exchanges/d or automated peritoneal dialysis (APD) nightly
Hemodynamics	Demanding — requires stable MAP	Gentle — ideal for shock	Intermediate	Very gentle — preserves residual function
Solute clearance	Highest — best for toxins / severe hyperK	Moderate (20–25 mL/kg/h effluent)	Intermediate	Lowest — not for acute toxicity
Volume removal	Rapid (up to 4 L per session)	Continuous, titrated to hemodynamics	Intermediate	Slow; limited in peritoneal scarring
Access	AVF > arteriovenous graft (AVG) > central venous catheter (CVC)	Large-bore CVC (usually IJ)	CVC	PD catheter (Tenckhoff) — 2 weeks to mature
Best for	<i>Stable patient, toxin, severe hyperK, fast removal</i>	<i>Shock, high fluid requirements, brain injury (↓ ICP shifts)</i>	<i>Intermediate — ICU without full CRRT</i>	<i>Home modality, preserved residual function, pediatric</i>

Vascular access — life-plan first

Right access, right patient, right time. Plan months in advance.

Access	Pros	Cons	Maturation / prep
arteriovenous fistula (AVF) (native)	Longest life, lowest infection, best flow	Not all patients are candidates; may fail to mature	Often needs months; refer around eGFR 15–20 or earlier if rapid decline
AVG (graft)	Faster maturation, salvageable for obese/vasculopath	Higher thrombosis, infection, shorter life	2–4 weeks
Tunneled CVC	Immediate use	Infection (central line-associated bloodstream infection (CLABSI)), thrombosis, central stenosis → ruins future AVF	Temporary bridge only
PD catheter	Home modality; no needles; gentler	Peritonitis risk; abdominal surgery / hernias contraindicate	2 weeks to mature; embedded or buried option

Vein preservation matters: NO PICCs, NO IVs in non-dominant arm once eGFR < 30. Central lines ruin future AVF sites.

Complications during & between treatments

Anticipate them — they're common and often preventable.

DURING DIALYSIS

Hypotension — most common. Causes: excessive UF, short session, cardiac dysfunction. Management: lower UF rate, cool dialysate, midodrine, reassess dry weight.

Muscle cramps — high UF rate; reduce goal.

Arrhythmias — sudden cardiac death risk elevated on MWF schedule (long interval).

Access dysfunction — low flow, recirculation; urgent US / declotting.

Dialyzer reaction — eosinophilia, bronchospasm (Type A); pre-rinse and alternate dialyzer.

BETWEEN TREATMENTS

Volume gain > 3–5% dry weight → review intake, Na⁺ diet

Missed dialysis → hyperkalemia, volume overload, uremia

Access infection → fever, redness; blood cultures + vanco

Calciphylaxis — painful skin ulcers; low-Ca dialysate, sodium thiosulfate, wound care

Amyloidosis, RLS, pruritus — chronic complications of long-term HD

Home dialysis — PD and home HD

Independence, preserved residual function, comparable outcomes.

PERITONEAL DIALYSIS

Modes: CAPD (4 manual exchanges/day) or APD (cyclor at night)

Prescription: dwell volume, # exchanges, dextrose concentration (1.5/2.5/4.25%)

Icodextrin for long dwell (fewer failures to UF)

Adequacy: dialysis adequacy (clearance $\times$ time / distribution volume) (Kt/V) ≥ 1.7 /week (weekly urea clearance)

Peritonitis: cloudy effluent, fever, abdominal pain; WBC > 100 with $> 50\%$ PMN; IP antibiotics

Contraindications: large hernias, recent abdominal surgery, scarring

HOME HEMODIALYSIS

Daily short (2–3 h $\times$ 5–6 $\times$ /wk) or nocturnal (6–8 h $\times$ 3–6 $\times$ /wk)

Better BP, LVH regression, phosphorus control (FHN trial signal)

Requires care partner, home space, training (4–6 weeks)

Vascular access often AVF with buttonhole cannulation

Lower QoL burden for many compared to in-center

Best for motivated patients with support at home

Landmark trials in dialysis

What changed, what didn't.

Trial	Year	Takeaway
HEMO	2002	Higher Kt/V or high-flux dialyzer did NOT improve mortality in maintenance HD.
ATN Trial	2008	Intensive CRRT (35 mL/kg/h) = standard (20–25) in AKI. Standard is sufficient.
IDEAL	2010	Early vs late RRT start in ESRD — no mortality difference; early = more complications.
AKIKI	2016	Delayed KRT safe in stage 3 AKI; 49% never needed dialysis.
STARRT-AKI	2020	Accelerated KRT did not reduce 90-d mortality; 38% of standard arm avoided KRT.
ELAIN	2016	Single-center signal favoring early KRT — not replicated in STARRT-AKI.
FHN Daily	2010	6× weekly HD improved LVH, BP — no mortality signal.
PIVOTAL	2019	Proactive IV iron (high-dose) in HD ↓ CV events, ↓ ESA need.

Common presentation mistakes

Avoid these when presenting a KRT consult.

AVOID

- Starting dialysis on BUN alone
- Not knowing ESRD schedule / dry weight
- Skipping medical optimization
- CRRT without checking hemodynamics
- PICCs / peripheral IVs in CKD
- No backup access plan

DO INSTEAD

- State the clinical problem, not BUN
- Pull schedule, access, dry weight, transplant
- Name what medical therapy failed
- IHD if stable; CRRT if on pressors
- Vein preservation — no BP/IV ordered in arm
- Every CVC needs a next-step plan

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 62-year-old man admitted to MICU with septic shock from pneumonia, on norepinephrine 0.4 µg/kg/min and vasopressin. Oliguric × 24 h, Cr rose from 1.2 to 4.6 over 72 h. K⁺ 6.8 (persistent despite med therapy), HCO₃ 14, pH 7.18, volume-overloaded with worsening oxygenation. No urgent toxin.

QUESTION: Does he meet criteria for KRT, and what modality?

Clinical case — answer

Key teaching points from the case.

ANSWER

Yes — multiple AEIOU indications (Acidosis, Electrolytes, Overload). Start CRRT given hemodynamic instability.

TEACHING

AEIOU triggers: A (pH < 7.15), E (life-threatening or refractory hyperkalemia, often $K^+ \geq 6.5$ or any dangerous ECG changes), O (pulmonary edema limiting oxygenation). CRRT preferred over IHD because he's on high-dose pressors; CRRT's gentler solute/fluid removal avoids further hemodynamic collapse. Standard dose 20-25 mL/kg/h effluent (ATN trial — more is not better). Large-bore CVC (usually IJ) for access. Goal is to bridge to renal recovery; ~60% of AKI-KRT patients recover kidney function, so this does NOT mean lifelong dialysis.

REFERENCES

Dialysis / KRT

Guidelines

KDIGO 2012 AKI Guideline (RRT section)
KDOQI 2019 Vascular Access Update
KDOQI 2015 Hemodialysis Adequacy
ISPD 2022 Peritonitis Guidelines
ASDIN and ASN Access & Modality statements

Landmark Trials

HEMO — NEJM 2002
ATN Trial — NEJM 2008
IDEAL — NEJM 2010
AKIKI — NEJM 2016
ELAIN — JAMA 2016
STARRT-AKI — NEJM 2020
FHN Daily — NEJM 2010
PIVOTAL — NEJM 2019

UpToDate (2026 access):

Kidney replacement therapy in AKI: indications and timing • Selection of modality for acute RRT • Overview of peritoneal dialysis • Intradialytic hypotension • Vascular access for hemodialysis